

Bhavya Singh

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EDUCATION

ABV-Indian Institute of Information Technology and Management (ABV-IIITM)
B.Tech in Information Technology + MBA • **Grade:** 7.74 CGPA

PROJECTS

Mesh-Agent

[\[View Project\]](#)

- Developed a comprehensive agent application engineered with a robust **Python backend** and deployed via **Docker containers** for scalable microservices architecture.
- Designed a multi-faceted web frontend utilizing custom **HTML/CSS templates** and responsive design to deliver a seamless user experience.
- Integrated core user interface modules including a dedicated real-time chat system, terminal access, and isolated workspace environments.
- Implemented configuration management dashboards alongside **CI/CD pipelines** using **GitHub Actions**.

AI-File-Organizer

[\[View Project\]](#)

- Created an intelligent file organization utility that leverages **Generative AI** and **Machine Learning** to automate local file management.
- Built a sleek graphical user interface (GUI) utilizing the **CustomTkinter** library to ensure an intuitive user experience.
- Integrated **OpenAI API** and **Ollama (Local LLMs)** models to perform semantic analysis and categorize files into predefined themes via **prompt engineering**.
- Utilized the **Pillow (PIL)** library for advanced image processing and metadata extraction, ensuring a robust and versatile application architecture.

Chat-With-PDF

[\[View Project\]](#)

- Engineered a robust Python application leveraging **Retrieval-Augmented Generation (RAG)** to facilitate interactive, conversational querying of PDF documents.
- Developed a custom **command-line interface (CLI)** to provide users with direct and efficient access to Natural Language Processing (NLP) features.
- Implemented specialized modules for document loading, **vector text embeddings**, and a core processing engine to handle complex data workflows.
- Automated the deployment pipeline utilizing GitHub Actions to seamlessly publish the Python package to **PyPI**.

Wiki-Links-Tree

[\[View Project\]](#)

- Designed a **Data Visualization** tool built in Python to parse and map Markdown-based documentation vaults using graph data structures.
- Utilized the **Pyvis** library to generate interactive network graphs representing the relational interconnections between various Wiki pages.
- Engineered the core logic within a dedicated module to extract internal links from files, optimizing parsing algorithms for large document vaults.

Knowledge-Graph-3D

[\[View Project\]](#)

- Constructed an interactive three-dimensional **Knowledge Graph** application to visualize complex relational databases and interconnected data.
- Developed a dedicated graphical user interface (GUI) to allow users to easily navigate, zoom, and interact with the 3D coordinate space.
- Implemented custom **Python scripting** to manage the dynamic generation and spatial positioning of data points.

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, C, JavaScript, HTML5, FastAPI

Libraries & Tools: scikit-learn, Pandas, NumPy, Docker, Git, CI/CD (GitHub Actions)

